



Department of Information Technology (Bsc.IT Semester III)

Python Programming Practical

Practical – 8

Roll No. :	Name:
Class :	Date/Time of Submission: 4-08-2026

Practical No. 8 - Write Algorithm, Code and Output

Q. 1. Write algorithm for the following:

a. Write a program to accept and pass radius to a function that returns area and circumference (using tuple).

Code:

```
import math
def circle_details(r):
    area=math.pi*r
    circum=2*math.pi*r
    return(area,circum)
radius=int(input("Enter the value : "))
result=circle_details(radius)

print("Area: " ,result[0])
print("Circumference : ",result[1])
print("Himanshu Shrivastav S037")
```

Output:

```
h/1 y chom14/geometry.py -
Enter the value : 12
Area:  37.69911184307752
Circumference :  75.39822368615503
Himanshu Shrivastav S037
```

b. Write a program to perform basic file operations on text files and binary files.

Code:

```

with open("FiletoRead.txt","w")as f:
    f.write("Hello The Furious Future of MVLU.")
with open("FiletoRead.txt","r")as f:
    content=f.read()
    print("Text File Content: ",content)
print("Himanshu S037")

```

For Binary Files:

```

with open("b_sample.bin","wb")as bf:
    data=b"Hello The Furious Future of MVLU Having Data In Binary File."
    bf.write(data)
with open("b_sample.bin","rb")as bf:
    b_content=bf.read()
    print("Binary File Content: ",b_content)
print("Himanshu S037")

```

Output:

```

Text File Content:  Hello The Furious Future of MVLU.
Himanshu S037

----- RESTART: C:/Users/mvlu/Desktop/new folder/file.py -----
Binary File Content:  b'Hello The Furious Future of MVLU Having Data In Binary F
ile.'
Himanshu S037

```

c. Write a Python program to read last n lines of a file.

Code:

```

def read_last_n_lines(filename, n):
    with open(filename, "r") as f:
        lines = f.readlines()
    return lines[-n:]

```

```
filename = "FiletoRead.txt"
```

```
n = int(input("Enter number of last lines to read: "))
```

```
last_lines = read_last_n_lines(filename, n)
print("\nLast ", n, "Lines: ")
for line in last_lines:
    print(line.strip())
print("Himanshu S037")
```

Output:

```
== RESTART: C:/Users/mvluc/Desktop/New folder/
Enter number of last lines to read: 2

Last 2 Lines:
Hello The Furious Future of MVLU.
Himanshu S037
```

Q. 2. What is file? What are its operating modes?

Q. 3. Write short note on file object attributes and file methods.